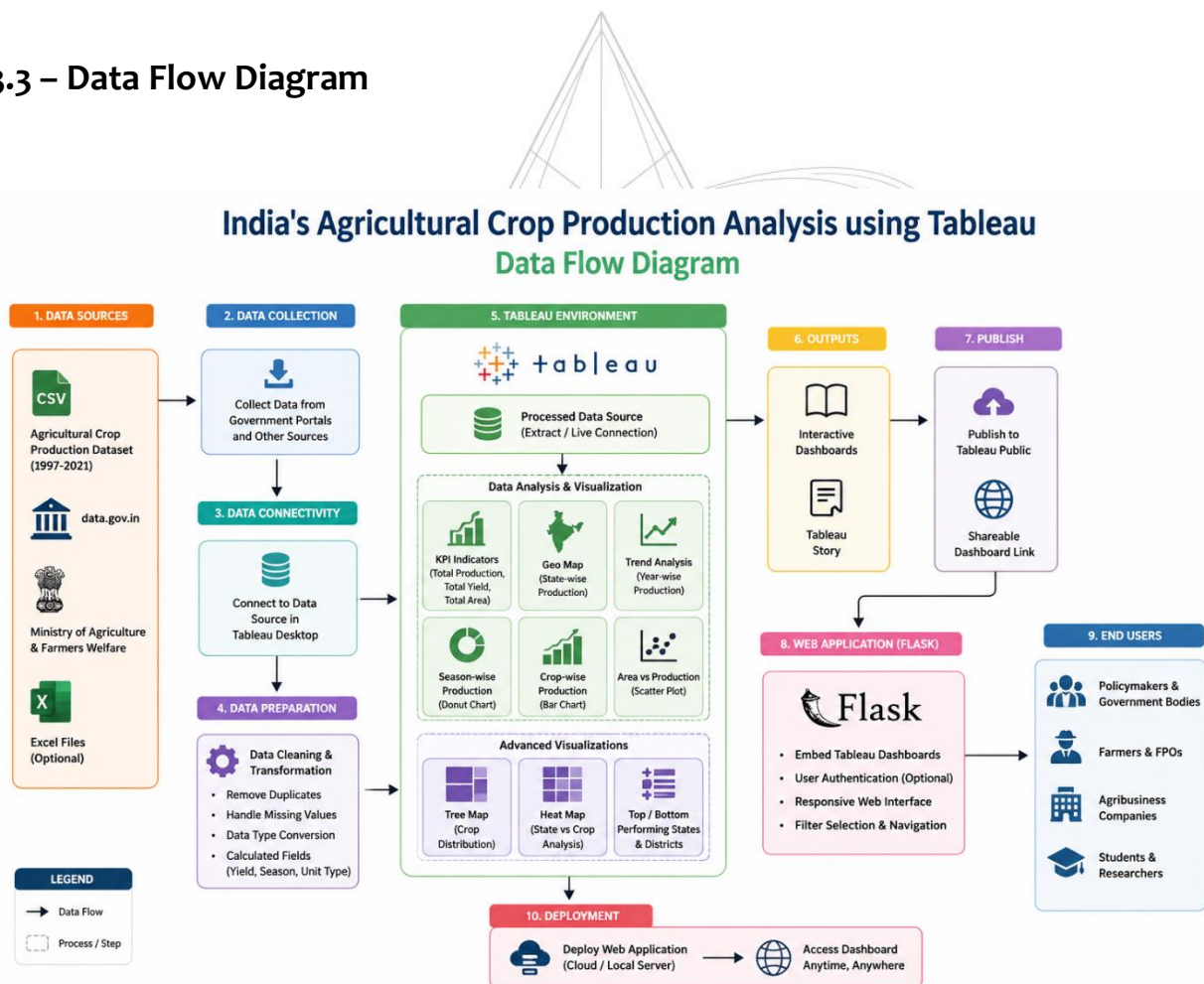


India's Agricultural Crop Production Analysis using Tableau

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	3 MARKS

3.3 – Data Flow Diagram



Data Collection

Collect crop production data from government websites like data.gov.in. Get CSV files with crop production, area and yield data for 1997-2021 across states and districts.

Data Cleaning & Transformation

Connect to the data in Tableau Desktop. Clean it. Remove records fix missing values convert data types and create new fields like Yield (Production divided by Area) Season categories and Unit Type.

Data Connectivity & KPI Creation

Link the data and create key indicators for Total Production, Yield and Area. Make charts like a donut chart for season-wise production a bar chart for crop-wise production and a scatter plot for Area vs Production.

Data Preparation for Advanced Analysis

Prepare data for charts by structuring it. Create Tree Maps for crop distribution Heat Maps for State vs Crop analysis. Find top and bottom performing states and districts.

Tableau Environment Setup

Set up Tableau with the processed data using Extract or Live Connection. Analyze the data. Create advanced charts like Tree Maps, Heat Maps and performance rankings.

Dashboard & Story Creation

Combine charts into dashboards with filters and drill-down capabilities. Create a Tableau Story to present findings in a format for stakeholders.

Web Application Integration (Flask)

Develop a Flask web application to embed Tableau dashboards. Add user authentication, filter selection and navigation features.

Publish & Share

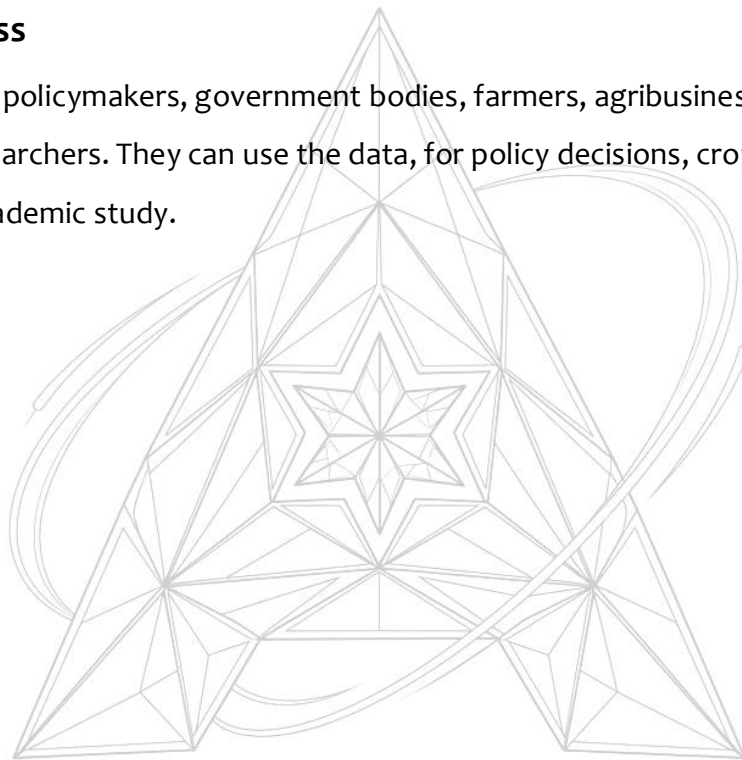
Publish dashboards to Tableau Public. Generate links to share with users and stakeholders.

Deployment

Deploy the web application on cloud platforms or local servers. Ensure dashboards are accessible anytime anywhere for policymakers, farmers, companies, students and researchers.

End User Access

Provide access to policymakers, government bodies, farmers, agribusiness companies, students and researchers. They can use the data, for policy decisions, crop planning, market strategies and academic study.



ALPHA
— INNOVATORS —

